Abdelaziz M. A. Ibrahim

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Egyptian Abdulaziz Ibrahim Abdelaziz Ibrahim

Profile

Graduate in Computer Science and Artificial Intelligence with a strong academic record, currently pursuing a Master's at the University of Jyväskylä, Finland. Experienced Teaching Assistant skilled in Oracle SQL, Python, and Deep Learning, with a focus on developing innovative solutions.

Education

08/2024 – present
Jväskylä, Finland

Master's Degree in Artificial Intelligence, *University of Jyväskylä*
Deep Learning, Cloud Computing, Linked Data

09/2016 – 05/2020
Helwan, Egypt

Bachelor of Computers and Information, *Helwan University*
Top of my class 4 years straight, GPA 3.83/4

Professional Experience

06/2022 – 09/2024
Helwan, Egypt

Teaching Assistant, *Helwan University*
Responsibilities: Teaching undergraduate courses, developing teaching materials, preparing and grading examinations
Courses Assisted: Natural Language Processing (NLP), Machine Learning (ML), Evolutionary Algorithms.

09/2021 – 06/2022
Cairo, Egypt

Database Specialist, *Armed Forces*

- Developed and maintained database reports, writing complex queries using **Oracle SQL.**
- Technical support using **.NET** framework, resolving issues and adding new features.

Skills

Programming Languages: Python, Java, C#

AI/ML tools: Keras, scikit-learn, OpenCV, DEAP, PyTesseract

Database Technologies: Oracle SQL, SPARQL, RDF

Web Development: .NET, Flask, Jersey

Other Skills: Knowledge Graphs, Evolutionary Algorithms (PSO, Genetic), TKinter GUI

Soft Skills: Sharp Memory, Persistent, Fast Learner, Adaptable

Languages

English: C1, IELTS Academic OVR: 8

Arabic: Native

Online Courses

Supervised Machine Learning: Regression and Classification

Advanced Learning Algorithms

Mathematics for Machine Learning: Linear Algebra

Introduction to Cloud Computing

Projects

Data Entry Automation System(Graduation Project),
Image Processing, OpenCV, Sklearn, Pytesseract, XlsxWriter

Digitization of printed paper to word, Excel, or text.

- **Table Extraction** (Printed/Handwritten)
- **Table Reconstruction**
- **Optical Character Recognition**
  - English and Arabic word and character segmentation
  - Creation of an Arabic Data Set for a specific Font

Multi-Agent System for Automated Literature Reviews, *Python, SPADE*

A multi-agent system in Python using the SPADE framework to automate and streamline the academic literature review process. The system features:
Specialized Intelligent Agents: Implemented distinct agents for query construction, paper searching (arXiv), relevance assessment, knowledge aggregation (fetching full-text content via Jina Reader API and HTML parsing), paper analysis (extracting methodology, findings, future work), and final report synthesis.
LLM Integration: Integrated Google's Gemini LLM for NLP tasks, evaluating paper relevance, and detailed content analysis.

Domain Specific RAG, *python, Streamlit, Langchain, FAISS*

a Retrieval Augmented Generation (RAG) application enabling users to interact with PDF documents through a conversational interface.

- **Interactive PDF Chat**
- **Contextual Conversation:** Maintains conversation history to understand follow-up questions and provide contextually relevant responses.
- **Information Retrieval:** a vector store (FAISS) for quick and relevant information retrieval from document embeddings.
- **Source Referencing**

Traffic Sign Classification Web APP, *Flask, Keras, EfficientNetV2*

Developed a real-time web application for classifying traffic signs into 58 distinct categories. Employed transfer learning using EfficientNetV2 model

Information Retrieval (text-based querying.), *python, Indexing, Tokenization, Vector space*

Creating a Search Engine on a collection of Agatha Christie's stories.

- **Tokenization and Text Preprocessing:** Cleaned and normalized text data.
- **Indexing:** Built an inverted index to map terms to document occurrences.
- **Search Mechanism:** Vector Space Model with TF-IDF weighting to rank and retrieve relevant documents based on user queries.

Vehicle Renting System, *java, OOP, (Class diagram, ERD, DFD, sequence diagram, etc...)*

A java desktop application for managing vehicle rentals.

- **Customer Module:** Search for available cars, view details, and rent vehicles.
- **Admin Module:** Add, update, and delete car listings and manage customer information.